



RELAZIONE PERIODICA

Mesi di novembre-dicembre 2025

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CUP progetto	I53C21000370006
Corso di dottorato	Applied Data Science and Artificial Intelligence (ADSAI) Università degli Studi di Trieste
Ciclo-anno accademico	XXXIX
Tutor scientifico	Monica Sugan

ATTIVITÀ DI RICERCA E FORMATIVE TUTORAGGIO E DIDATTICA CONFERENZE RISULTATI DELLA RICERCA ALTRO

Modernized OGS catalog evaluation and deployment. The changes added EQTransformer/PhaseNet model configurations, reorganized velocity/magnitude data paths, corrected comparison labels and metrics, introduced event/pick time-window grouping, and optimized graph matching through vectorized conversions, station indexing, and temporal pre-filtering. Large generated-result archives were refreshed and obsolete diagnostic images removed. The merge consolidated the two parallel November branches without adding distinct functionality.

Conferences:

K. Tanaka Hernández (2025). Artificial Intelligence (AI) and Machine Learning (ML) on Big Data Seismology. 24th Mexican International Conference on Artificial Intelligence, MICA 2025 Doctoral Consortium, 3 - 7 November, 2025, Guanajuato, Mexico.

REFERENCES:

Mousavi, S. M., Ellsworth, W. L., Zhu, W., Chuang, L. Y., & Beroza, G. C. (2020). Earthquake transformer—An attentive deep-learning model for simultaneous earthquake detection and phase picking. *Nature Communications*, **11**(1), 1–12. <https://doi.org/10.1038/s41467-020-17591-w>

Münchmeyer, J. (2024). PyOcto: A high-throughput seismic phase associator. *Seismica*, **3**(1). <https://doi.org/10.26443/seismica.v3i1.1130>

Woollam, J., J. Münchmeyer, F. Tilmann, A. Rietbrock, D. Lange, T. Bornstein, T. Diehl, C. Giunchi, F. Haslinger, D. Jozinović, et al. (2022). SeisBench—A Toolbox for Machine Learning in Seismology, *Seismol. Res. Lett.* **93**, 1695–1709, doi: 10.1785/0220210324.

Zhu, W., & Beroza, G. C. (2019). Phasenet: A deep-neural-network-based seismic arrival-time picking method. *Geophysical Journal International*, **216**(1), 261–273. <https://doi.org/10.1093/gji/ggy423>

Zhu, W., McBrearty, I. W., Mousavi, S.M., Ellsworth, W. L., & Beroza, G. C.(2022). Earthquake phase association using a Bayesian Gaussian Mixture Model. *Journal of Geophysical Research: Solid Earth*, **127**, e2021JB023249. <https://doi.org/10.1029/2021JB023249>

Dottorando	Tutor scientifico
<i>Firma</i>	<i>Firma</i> 

Data: 20 gennaio 2025